Majorana Modes in the Machine

Week 7: From ED Validation to a Real VQE Sweep

Dobromir Stoev
Edgar Harutyunyan
Guilherme Schewtschik
Jaskaran Singh
Zhengyi Liu
Zhenming Shi

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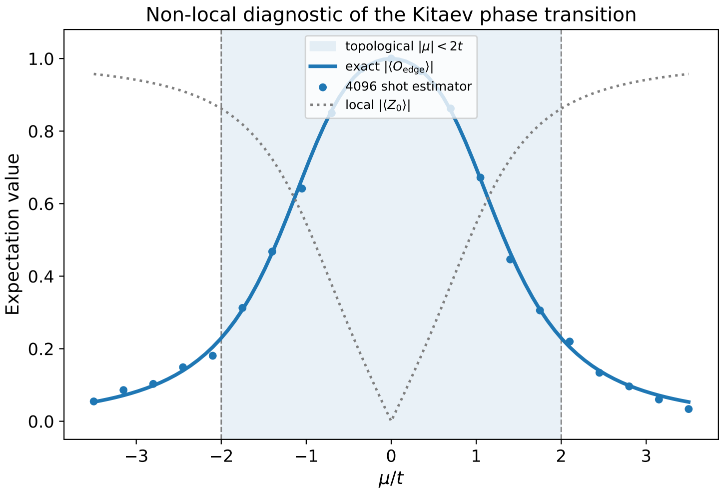
Week 7 Objective

Milestone

Replace the exact-diagonalization state preparation used in Week 6 with a parity-constrained VQE preparation at every point of the μ sweep.

- Keep the same Hamiltonian, edge-string observable, and shot-measurement protocol.
- Reuse the existing $L = 4$ hardware-efficient ansatz and optimizer workflow.
- Validate every VQE state against ED before introducing hardware noise.
- Produce a VQE-prepared phase sweep that can become the noiseless Block 4 baseline.

Where Week 6 Stops



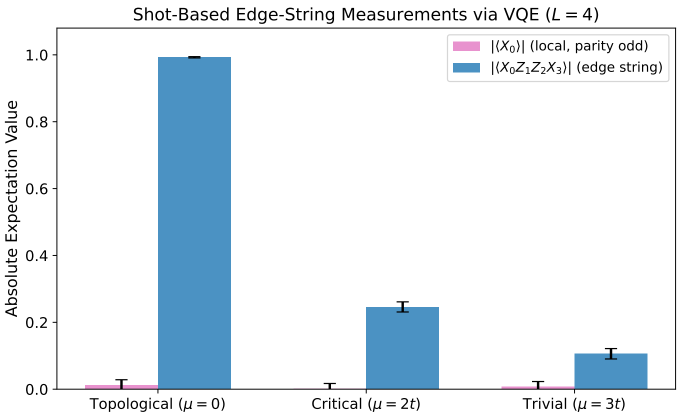
Already established

- Sweep target: $\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle$
- ED parity-gap cross-check
- Synthetic shot estimator

Remaining gap

- Every plotted state is prepared by ED, not VQE.

Existing VQE Baseline

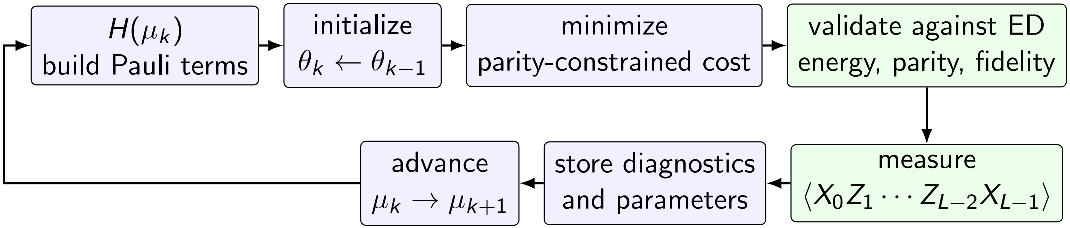


Existing components:

- `EfficientSU2`: R_Y + linear CNOTs
- Aer estimator + multi-start COBYLA
- ED fidelity checks
- Shot measurements

Current scope: three representative μ values.

The VQE Sweep Workflow



Key engineering change

The optimizer is no longer restarted independently at every μ; neighboring ground states seed one another.

Why Warm Starts Matter

Independent random starts

- Repeats expensive optimization work
- Can converge to different local minima
- Makes the sweep noisy even before shot noise

Continuation in μ

- Use $\theta^*(\mu_k)$ to initialize μ_{k+1}
- Exploits smooth ground-state evolution away from criticality
- Random restarts remain a fallback when validation fails

Expected difficulty

Near $|\mu| = 2t$, the bulk gap closes and the optimal state changes fastest; this is where warm starts and fidelity checks are most important.

Parity Must Be Controlled

$$P = \prod_{j=0}^{L-1} Z_j, \qquad [H, P] = 0$$

Problem

- The topological phase has nearly degenerate parity sectors.
- Energy minimization alone can return inconsistent sectors across the sweep.
- A sector jump can look like a discontinuity in the observable.

Week 7 strategy

$$C(\theta) = \langle H \rangle + \lambda\big(1 - \langle P \rangle\big)^2$$

- Target even parity consistently.
- Record $\langle P \rangle$ at every point.
- Reject states outside the parity tolerance.

Validation Dashboard

A low VQE energy alone is not enough. Each point must pass four checks:

Metric	Question	Use
Energy error	Is E_{VQE} near E_{ED}?	Detect optimizer failure
Parity	Is $\langle P \rangle \approx +1$?	Keep one sector across μ
Subspace fidelity	Does VQE overlap the correct low-energy space?	Handle near-degeneracy correctly
String error	Matches the ED edge string?	Validate the quantity used in the phase sweep

Pass criterion

Failed points trigger deeper ansatz repetitions, additional optimizer iterations, or random-restart recovery.

Implementation Shape

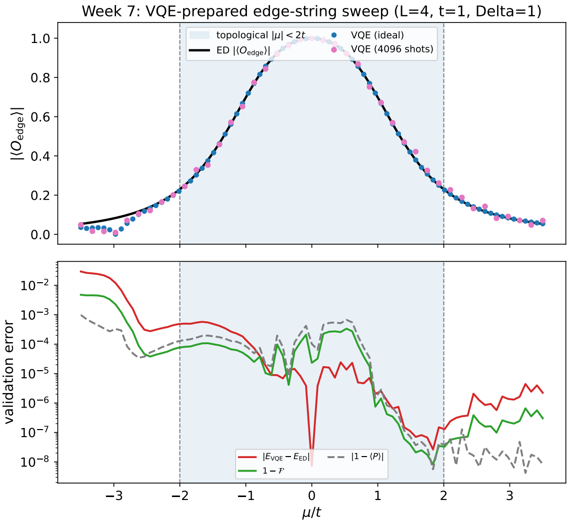
```
theta = initial_parameters()
for mu in mu_values:
    H = qiskit_hamiltonian(L, t, mu, delta)
    theta = optimize(parity_cost(H), initial=theta)

    diagnostics = validate(theta, H, ed_reference(mu))
    if not diagnostics.pass_all:
        theta = recover_with_restarts(H)

    string_value = measure_edge_string(theta, shots=4096)
    save(mu, theta, diagnostics, string_value)
```

- The sweep must save optimizer parameters and diagnostics, not only the final observable.
- Reproducible checkpoints make failed regions debuggable and reusable for later noise studies.

Week 7 Result: VQE-Prepared Sweep

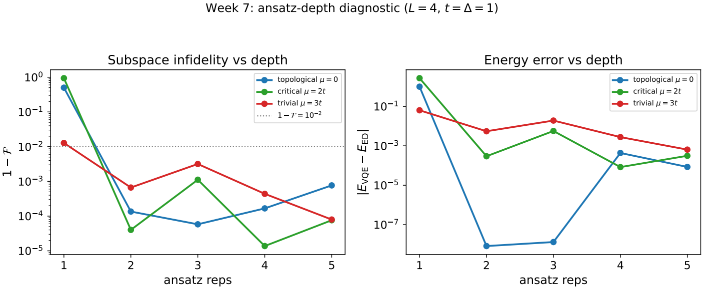


Top: edge string $|\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle|$ from parity-constrained VQE (ideal and shot-based) tracks the ED baseline across the whole μ grid.

Bottom: per-point validation — energy error, infidelity $1 - \mathcal{F}$, and parity deviation $|1 - \langle P \rangle|$. Crosses mark points needing random-restart recovery.

Warm-start continuation keeps every point in the even sector.

Ansatz Depth: How Deep Is Enough?



Multi-start VQE at three representative μ, scanning ansatz reps.

- 1 rep (8 params) is too shallow, worst at $\mu = 2t$ (fidelity ~ 0.06).
- 2+ reps already reach the even-parity ground space.
- Criticality is the hardest point, exactly where the bulk gap closes.

This justifies the `reps`$= 3$ used in the sweep.

Definition of Done for Week 7

- A scripted $L = 4$, $t = \Delta = 1$ VQE sweep over the same μ grid as Week 6.
- Consistent even-parity preparation across the sweep.
- ED comparison for energy, parity, subspace fidelity, and edge-string expectation.
- Shot-based VQE edge-string curve plotted against the ED baseline.
- Optimization cost and failure/recovery statistics recorded per μ point.

Boundary to Block 4

Only after the ideal VQE sweep passes validation should depolarizing, readout, and relaxation noise be added.

Next: Block 4 NISQ Reality Check

- Freeze the validated ideal VQE sweep as the noiseless reference.
- Apply realistic gate, readout, and relaxation noise to preparation and measurement circuits.
- Separate optimizer degradation from measurement degradation.
- Compare raw and mitigated edge-string curves.
- Quantify when the inferred phase boundary becomes unreliable.

Conceptual progression

Week 6 validated the diagnostic with ED; Week 7 validates quantum state preparation; Block 4 tests whether the full workflow survives realistic noise.